

## **Firewall Configuration**

### **Overview**

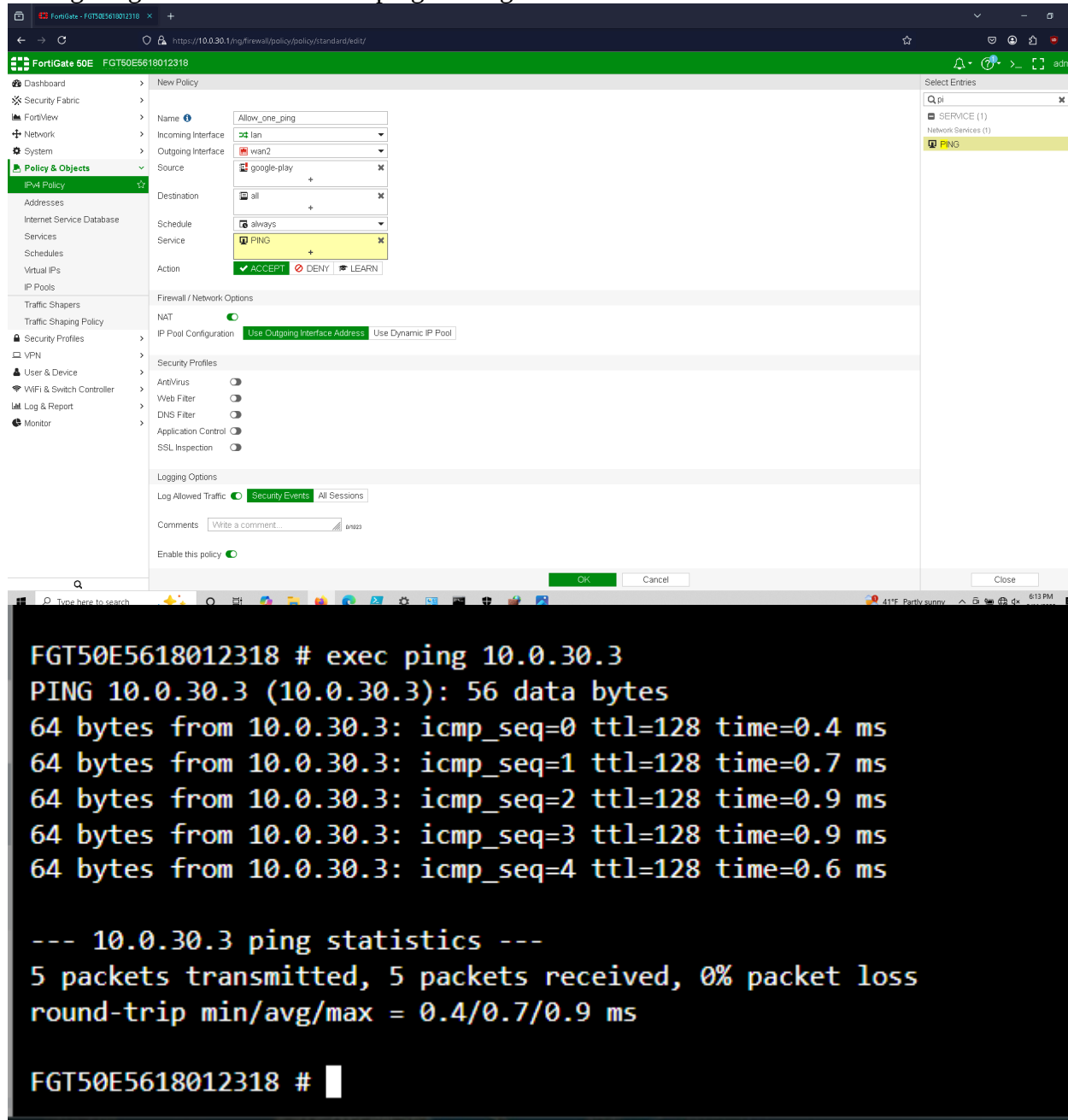
This section documents the configuration of the Fortinet firewall within the cyber range environment. The firewall was configured to manage network traffic, enforce security policies, and control access between internal and external systems.

## Allowing Outbound Traffic

The firewall was configured to allow internal systems to communicate with external networks.

- Configured policies to allow outbound traffic
- Verified connectivity by successfully pinging external devices from within the firewall

Configuring the firewall to allow pings through to internal:



The screenshot displays the FortiGate 50E configuration interface for a new policy named 'Allow\_one\_ping'. The policy is configured with the following settings:

- Name:** Allow\_one\_ping
- Incoming Interface:** lan
- Outgoing Interface:** wan2
- Source:** google-play
- Destination:** all
- Schedule:** always
- Service:** PING
- Action:** ACCEPT (checked), DENY, LEARN

The Firewall / Network Options section shows NAT enabled and IP Pool Configuration set to 'Use Outgoing Interface Address' and 'Use Dynamic IP Pool'. The Security Profiles section shows AntiVirus, Web Filter, DNS Filter, Application Control, and SSL Inspection all disabled. The Logging Options section shows Log Allowed Traffic enabled, Security Events selected, and All Sessions selected. The Comments field is empty, and the Enable this policy checkbox is checked.

Below the configuration interface, a terminal window shows the command 'exec ping 10.0.30.3' and its output, confirming successful connectivity:

```
FGT50E5618012318 # exec ping 10.0.30.3
PING 10.0.30.3 (10.0.30.3): 56 data bytes
64 bytes from 10.0.30.3: icmp_seq=0 ttl=128 time=0.4 ms
64 bytes from 10.0.30.3: icmp_seq=1 ttl=128 time=0.7 ms
64 bytes from 10.0.30.3: icmp_seq=2 ttl=128 time=0.9 ms
64 bytes from 10.0.30.3: icmp_seq=3 ttl=128 time=0.9 ms
64 bytes from 10.0.30.3: icmp_seq=4 ttl=128 time=0.6 ms

--- 10.0.30.3 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0.4/0.7/0.9 ms

FGT50E5618012318 #
```

## Pinging from inside the firewall to outside devices:

FortiGate 60E FGT50E5618012318

Dashboard Edit Interface

Security Fabric

FortiView

Network

Interfaces

DNS

SD-WAN

SD-WAN Status Check

SD-WAN Rules

Static Routes

System

Policy & Objects

Security Profiles

VPN

User & Device

WiFi & Switch Controller

Log & Report

Monitor

Interface Name: wan1 (70 4C A5 B8 84 F3)

Alias:

Link Status: Down

Type: Physical Interface

Role: WAN

Estimated Bandwidth: 0 Kbps Upstream 0 Kbps Downstream

Addressing mode: Manual DHCP PPPoE

IP/Network Mask: 10.0.200.13/255.255.255.0

Administrative Access:

IPv4: ☐ HTTPS ☐ HTTP ☐ PING ☒ FMS-Access ☐ CAPWAP

☐ SSH ☐ SNMP ☐ FTM ☐ RADIUS Accounting

☐ FortiTelemetry

Miscellaneous:

Scan Outgoing Connections to Botnet Sites:

Secondary IP Address:

Status:

Comments:

Interface State:

OK Cancel

```
C:\Users\linko>ping 10.0.30.1

Pinging 10.0.30.1 with 32 bytes of data:
Reply from 10.0.30.1: bytes=32 time<1ms TTL=255
Reply from 10.0.30.1: bytes=32 time<1ms TTL=255
Reply from 10.0.30.1: bytes=32 time<1ms TTL=255
Reply from 10.0.30.1: bytes=32 time=1ms TTL=255

Ping statistics for 10.0.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\linko>
```

## Restricting Access to the Firewall

Security measures were implemented to reduce exposure:

- Disabled ping responses on the firewall
- Prevented external systems from directly interacting with the firewall

This helps reduce the attack surface and limits unnecessary visibility of the device.

Preventing unauthorized IPs from pinging firewall:

New Policy

|                    |                                                                                                                    |
|--------------------|--------------------------------------------------------------------------------------------------------------------|
| Name               | Bad IP                                                                                                             |
| Incoming Interface | wan1                                                                                                               |
| Outgoing Interface | wan1                                                                                                               |
| Source             | Bad IP                                                                                                             |
|                    | +                                                                                                                  |
| Destination        | all                                                                                                                |
|                    | +                                                                                                                  |
| Schedule           | always                                                                                                             |
| Service            | ALL                                                                                                                |
|                    | +                                                                                                                  |
| Action             | <input checked="" type="checkbox"/> ACCEPT <input checked="" type="checkbox"/> DENY <input type="checkbox"/> LEARN |

☐ Log Violation Traffic

Comments  0/1023

Enable this policy ☒

OK Cancel

## Virtual IP (VIP) Configuration

Virtual IPs were created to allow controlled access to internal services:

- Configured port forwarding from external to internal systems
- Enabled access to services such as FTP and SSH
- Grouped VIPs for easier management

Making Virtual IPs:

The screenshot displays the FortiGate 50E configuration interface. The top bar shows the device name 'FGT50E5618012318' and the user 'admin'. The left sidebar contains navigation menus for Dashboard, Security Fabric, FortiView, Network, System, Policy & Objects, Virtual IPs, IP Pools, Traffic Shapers, Security Profiles, VPN, User & Device, WiFi & Switch Controller, Log & Report, and Monitor.

The main content area shows the 'Virtual IPs' configuration page. It lists several Virtual IP configurations under the 'IPv4 Virtual IP (4)' group:

| Name                       | Details                                       | Interface | Services | Ref. |
|----------------------------|-----------------------------------------------|-----------|----------|------|
| <b>IPv4 Virtual IP (4)</b> |                                               |           |          |      |
| FTP Server IP              | 10.0.30.13 --> 10.0.200.13 (TCP: 21 --> 21)   | wan1      |          | 1    |
| SSH Server IP              | 10.0.30.23 --> 10.0.200.23 (TCP: 22 --> 22)   | wan1      |          | 1    |
| Web Server IP              | 10.0.30.33 --> 10.0.200.33 (TCP: 80 --> 80)   | wan1      |          | 1    |
| ICMP IP                    | 10.0.30.43 --> 10.0.200.43 (TCP: 445 --> 445) | wan1      |          | 1    |

Below this, there are two 'IPv4 Virtual IP Group (2)' entries:

| Name                             | Details                                     | Interface | Services | Ref. |
|----------------------------------|---------------------------------------------|-----------|----------|------|
| <b>IPv4 Virtual IP Group (2)</b> |                                             |           |          |      |
| VIP Group ICMP                   | ICMP IP                                     | wan1      |          | 1    |
| VIP Group Other Services         | FTP Server IP, SSH Server IP, Web Server IP | wan1      |          | 1    |

The bottom section shows the 'New Virtual IP' configuration form. The 'Name' field is 'FTP Server IP'. The 'Interface' is 'wan1'. The 'Type' is 'Static NAT'. The 'External IP Address/Range' is '10.0.30.13'. The 'Mapped IP Address/Range' is '10.0.200.13'. The 'Optional Filters' section is expanded, showing 'Port Forwarding' is enabled. The 'Protocol' is 'TCP'. The 'External Service Port' is '21'. The 'Map to Port' is '21'. The 'OK' and 'Cancel' buttons are at the bottom right.

Outcome

The firewall was successfully configured to:

- Allow controlled outbound communication
- Restrict direct access to the firewall

- Block unwanted traffic
- Provide secure access to internal services through VIPs

Firewall complete:

| <div> <div>+ Create New</div> <div>Edit</div> <div>Delete</div> <div>Policy Lookup</div> <div>Search</div> <div>Q</div> </div> <div>Interface Pair View</div> <div>By Sequence</div> |                       |             |                          |          |         |        |         |                   |          |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------|--------------------------|----------|---------|--------|---------|-------------------|----------|--|
| ID                                                                                                                                                                                   | Name                  | Source      | Destination              | Schedule | Service | Action | NAT     | Security Profiles | Log      |  |
| lan → wan1 1                                                                                                                                                                         |                       |             |                          |          |         |        |         |                   |          |  |
| 2                                                                                                                                                                                    | internet_access       | all         | all                      | always   | ALL     | ACCEPT | Enabled |                   | UTM      |  |
| lan → wan2 1                                                                                                                                                                         |                       |             |                          |          |         |        |         |                   |          |  |
| 4                                                                                                                                                                                    | Allow_one_ping        | google-play | all                      | always   | PING    | ACCEPT | Enabled |                   | UTM      |  |
| wan1 → lan 2                                                                                                                                                                         |                       |             |                          |          |         |        |         |                   |          |  |
| 7                                                                                                                                                                                    | VIP Policy Ping ICMP  | all         | VIP Group ICMP           | always   | PING    | ACCEPT | Enabled |                   | UTM      |  |
| 6                                                                                                                                                                                    | VIP Group Ping Others | all         | VIP Group Other Services | always   | PING    | ACCEPT | Enabled |                   | UTM      |  |
| wan1 → wan1 1                                                                                                                                                                        |                       |             |                          |          |         |        |         |                   |          |  |
| 5                                                                                                                                                                                    | Bad IP                | Bad IP      | all                      | always   | ALL     | DENY   |         |                   | Disabled |  |
| wan2 → lan 1                                                                                                                                                                         |                       |             |                          |          |         |        |         |                   |          |  |
| 3                                                                                                                                                                                    | idk man               | all         | all                      | always   | ALL     | ACCEPT | Enabled | +                 | UTM      |  |
| Implicit 1                                                                                                                                                                           |                       |             |                          |          |         |        |         |                   |          |  |
| 0                                                                                                                                                                                    | Implicit Deny         | all         | all                      | always   | ALL     | DENY   |         |                   | Disabled |  |